



National Load Despatch Centre
राष्ट्रीय भार प्रेषण केंद्र
GRID CONTROLLER OF INDIA LIMITED
ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड

(Government of India Enterprise/ भारत सरकार का उद्यम)
B-9, QUTUB INSTITUTIONAL AREA, KATWARIA SARAI, NEW DELHI -110016
बी-9, कुतुब इन्स्टीट्यूशनल एरिया, कटवारिया सराये, न्यू दिल्ली-110016

Ref: POSOCO/NLDC/SO/Daily PSP Report

दिनांक: 24.07.2024

To,

- कार्यकारी निदेशक, पू.क्षे.भा.प्रे.के., 14, गोल्फ क्लब रोड, कोलकाता - 700033
Executive Director, ERLDC, 14 Golf Club Road, Tollygunge, Kolkata, 700033
- कार्यकारी निदेशक, ऊ.क्षे.भा.प्रे.के., 18/ए, शाहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली - 110016
Executive Director, NRLDC, 18-A, Shaheed Jeet Singh Marg, Katwaria Sarai, New Delhi - 110016
- कार्यकारी निदेशक, प.क्षे.भा.प्रे.के., एफ3, एम आई डी सी क्षेत्र, अंधेरी, मुंबई - 400093
Executive Director, WRLDC, F-3, M.I.D.C. Area, Marol, Andheri (East), Mumbai-400093
- कार्यकारी निदेशक, ऊ.पू.क्षे.भा.प्रे.के., डोंगतेह, लोअर नोंग्रह, लापलंग, शिलोंग - 793006
Executive Director, NERLDC, Dongteih, Lower Nongrah, Lapalang, Shillong - 793006, Meghalaya
- कार्यकारी निदेशक, द.क्षे.भा.प्रे.के., 29, रेस कोर्स क्रॉस रोड, बंगलुरु - 560009
Executive Director, SRLDC, 29, Race Course Cross Road, Bangalore-560009

Sub: Daily PSP Report for the date 23.07.2024.

महोदय/Dear Sir,

आईईजीसी-2010 की धारा स.-5.5.1 के प्रावधान के अनुसार, दिनांक 23-जुलाई-2024 की अखिल भारतीय प्रणाली की दैनिक ग्रिड निष्पादन रिपोर्ट रांभांप्रेके की वेबसाइट पर उपलब्ध है।

As per article 5.5.1 of the Indian Electricity Grid Code, the daily report pertaining power supply position of All India Power System for the date 23.07.2024, is available at the NLDC website.

धन्यवाद,
Thanks

Report for previous day

Date of Reporting: 24-Jul-2024

A. Power Supply Position at All India and Regional level

	NR	WR	SR	ER	NER	TOTAL
Demand Met during Evening Peak hrs(MW) (at 20:00 hrs; from RLDCs)	82312	55757	45079	28281	3523	214952
Peak Shortage (MW)	195	0	0	0	0	195
Energy Met (MU)	1880	1315	1069	618	68	4949
Hydro Gen (MU)	395	32	114	79	36	656
Wind Gen (MU)	16	215	297	-	-	528
Solar Gen (MU)*	152.06	50.02	87.55	1.25	1.10	292
Energy Shortage (MU)	1.18	0.00	0.00	0.08	0.00	1.26
Maximum Demand Met During the Day (MW) (From NLDC SCADA)	83834	57240	50321	29299	3526	215216
Time Of Maximum Demand Met	22:20	19:27	10:22	22:49	19:52	19:43

B. Frequency Profile (%)

Region	FVI	< 49.7	49.7 - 49.8	49.8 - 49.9	< 49.9	49.9 - 50.05	> 50.05
All India	0.024	0.00	0.00	3.26	3.26	80.62	16.12

C. Power Supply Position in States

Region	States	Max.Demand Met during the day (MW)	Shortage during maximum Demand (MW)	Energy Met (MU)	Drawal Schedule (MU)	OD(+)/UD(-) (MU)	Max OD (MW)	Energy Shortage (MU)
NR	Punjab	15565	0	360.4	220.9	-1.3	188	0.00
	Haryana	13650	0	283.9	210.9	-0.6	277	0.23
	Rajasthan	15407	0	331.8	116.5	3.1	910	0.36
	Delhi	7095	0	147.2	124.1	-1.1	228	0.00
	UP	29078	0	605.3	285.8	-1.0	596	0.00
	Uttarakhand	2396	0	52.0	26.2	0.7	179	0.59
	HP	1732	0	37.6	0.2	0.4	90	0.00
	J&K(UT) & Ladakh(UT)	2611	0	49.8	30.9	-6.8	215	0.00
	Chandigarh	390	0	7.9	8.1	-0.1	30	0.00
WR	Railways_NR ISTS	218	0	4.2	3.8	0.5	83	0.00
	Chhattisgarh	4936	0	105.9	62.1	-2.1	291	0.00
	Gujarat	18510	0	409.2	139.3	-0.5	846	0.00
	MP	11188	0	242.7	137.7	-3.8	451	0.00
	Maharashtra	22391	0	485.8	152.2	-6.4	621	0.00
	Goa	583	0	12.3	12.7	-0.6	114	0.00
	DNHDDPDCL	1270	0	29.5	29.9	-0.4	41	0.00
	AMNSIL	756	0	16.5	5.8	-0.4	300	0.00
	BALCO	528	0	12.6	12.7	-0.1	9	0.00
SR	Andhra Pradesh	9372	0	198.0	25.3	-0.4	467	0.00
	Telangana	11303	0	215.3	131.3	-0.6	606	0.00
	Karnataka	10438	0	192.6	-19.6	-1.4	483	0.00
	Kerala	3967	0	78.8	44.3	1.0	391	0.00
	Tamil Nadu	17340	0	373.7	159.6	3.2	1049	0.00
	Puducherry	494	0	10.5	10.4	-0.2	31	0.00
	Bihar	7588	0	159.0	154.2	-1.4	310	0.08
ER	DVC	3471	0	74.2	-38.4	-0.5	575	0.00
	Jharkhand	2190	0	44.5	33.5	-1.0	193	0.00
	Odisha	5311	0	110.1	31.1	-1.8	418	0.00
	West Bengal	10682	0	228.5	87.9	-1.4	301	0.00
	Sikkim	86	0	1.4	1.5	-0.2	28	0.00
	Railways_ER ISTS	17	0	0.0	0.2	-0.2	0	0.00
NER	Arunachal Pradesh	157	0	2.7	2.7	-0.6	23	0.00
	Assam	2402	0	46.3	37.9	0.8	126	0.00
	Manipur	175	0	2.5	2.7	-0.3	59	0.00
	Meghalaya	310	0	5.4	-0.7	0.0	78	0.00
	Mizoram	122	0	2.0	0.7	-0.1	51	0.00
	Nagaland	188	0	2.9	2.9	-0.4	18	0.00
	Tripura	304	0	5.7	5.6	-0.1	33	0.00

D. Transnational Exchanges (MU) - Import(+ve)/Export(-ve)

	Bhutan	Nepal	Bangladesh	Godda -> Bangladesh
Actual (MU)	42.0	16.0	-23.9	-22.3
Day Peak (MW)	1959.0	792.8	-1010.0	-1288.0

E. Import/Export by Regions (in MU) - Import(+ve)/Export(-ve); OD(+)/UD(-)

	NR	WR	SR	ER	NER	TOTAL
Schedule(MU)	444.4	-305.1	-54.5	-85.3	0.5	0.0
Actual(MU)	429.1	-305.9	-39.6	-90.8	-1.1	-8.2
O/D/U/D(MU)	-15.3	-0.8	14.9	-5.5	-1.6	-8.2

F. Generation Outage(MW)

	NR	WR	SR	ER	NER	TOTAL	% Share
Central Sector	3262	12933	6288	4716	280	27478	51
State Sector	3286	11970	9673	974	157	26059	49
Total	6548	24902	15961	5690	437	53538	100

G. Sourcewise generation (Gross) (MU)

	NR	WR	SR	ER	NER	All India	% Share
Coal	889	1328	519	694	14	3444	65
Lignite	29	12	52	0	0	93	2
Hydro	395	32	114	79	36	656	12
Nuclear	30	66	76	0	0	172	3
Gas, Naptha & Diesel	33	44	5	0	23	106	2
RES (Wind, Solar, Biomass & Others)	176	266	409	3	1	855	16
Total	1552	1748	1175	776	74	5326	100

Share of RES in total generation (%)	11.34	15.22	34.77	0.44	1.49	16.05
Share of Non-fossil fuel (Hydro,Nuclear and RES) in total generation(%)	38.72	20.85	50.92	10.56	50.09	31.60

H. All India Demand Diversity Factor

Based on Regional Max Demands	1.041
Based on State Max Demands	1.088

I. All India Peak Demand and shortage at Solar and Non-Solar Hour

	Max Demand Met(MW)	Time	Shortage(MW)
Solar hr	214638	10:22	0
Non-Solar hr	215216	19:43	75

Diversity factor = Sum of regional or state maximum demands / All India maximum demand

**Note: All generation MU figures are gross

***Godda (Jharkhand) -> Bangladesh power exchange is through the radial connection (isolated from Indian Grid)

Solar Hours -> 07:00 to 17:00hrs and rest are Non-Solar Hours

*Source: RLDCs for solar connected to ISTS; SLDCs for embedded solar. Limited visibility of embedded solar data.

Executive Director-NLDC

INTER-REGIONAL EXCHANGES

Import=(+ve) /Export =(-ve) for NET (MU)

Date of Reporting: 24-Jul-2024

Sl No	Voltage Level	Line Details	No. of Circuit	Max Import (MW)	Max Export (MW)	Import (MU)	Export (MU)	NET (MU)
Import/Export of ER (With NR)								
1	HVDC	ALIPURDUAR-AGRA	2	0	701	0.0	17.1	-17.1
2	HVDC	PUSAULI B/B	-	0	95	0.0	2.4	-2.4
3	765 kV	GAYA-VARANASI	2	804	556	0.0	5.1	-5.1
4	765 kV	SASARAM-FATEHPUR	1	369	265	0.0	2.0	-2.0
5	765 kV	GAYA-BALIA	1	0	818	0.0	13.3	-13.3
6	400 kV	PUSAULI-VARANASI	1	0	126	0.0	1.5	-1.5
7	400 kV	PUSAULI -ALLAHABAD	1	32	57	0.0	0.9	-0.9
8	400 kV	MUZAFFARPUR-GORAKHPUR	2	0	716	0.0	11.8	-11.8
9	400 kV	PATNA-BALIA	2	119	663	0.0	10.9	-10.9
10	400 kV	NAUBATPUR-BALIA	2	102	373	0.0	2.6	-2.6
11	400 kV	BIHARSHARIFF-BALIA	2	289	127	0.0	0.5	-0.5
12	400 kV	MOTIHARI-GORAKHPUR	2	29	396	0.0	6.1	-6.1
13	400 kV	BIHARSHARIFF-VARANASI	2	0	0	0.0	0.0	0.0
14	220 kV	SAHUPURI-KARAMNASA	1	0	85	0.0	1.5	-1.5
15	132 kV	NAGAR UNTARI-RIHAND	1	0	0	0.0	0.0	0.0
16	132 kV	GARWAH-RIHAND	1	30	0	0.6	0.0	0.6
17	132 kV	KARMANASA-SAHUPURI	1	0	0	0.0	0.0	0.0
18	132 kV	KARMANASA-CHANDAULI	1	0	0	0.0	0.0	0.0
ER-NR						0.6	75.5	-74.9
Import/Export of ER (With WR)								
1	765 kV	JHARSUGUDA-DHARAMJAIGARH	4	350	1072	0.0	9.9	-9.9
2	765 kV	NEW RANCHI-DHARAMJAIGARH	2	1942	0	29.3	0.0	29.3
3	765 kV	JHARSUGUDA-DURG	2	100	476	0.0	3.4	-3.4
4	400 kV	JHARSUGUDA-RAIGARH	4	24	426	0.0	5.3	-5.3
5	400 kV	RANCHI-SIPAT	2	498	46	5.6	0.0	5.6
6	220 kV	BUDHIPADAR-RAIGARH	1	29	71	0.0	0.4	-0.4
7	220 kV	BUDHIPADAR-KORBA	2	25	71	0.0	0.5	-0.5
ER-WR						34.9	19.6	15.4
Import/Export of ER (With SR)								
1	HVDC	JEYPORE-GAZUWAKA B/B	2	108	0	2.5	0.0	2.5
2	HVDC	TALCHER-KOLAR BIPOLE	2	0	791	0.0	19.3	-19.3
3	765 kV	ANGUL-SRIKAKULAM	2	0	2365	0.0	31.5	-31.5
4	400 kV	TALCHER-I/C	2	539	23	10.7	0.0	10.7
5	220 kV	BALIMELA-UPPER-SILERRU	1	0	0	0.0	0.0	0.0
ER-SR						2.5	50.8	-48.3
Import/Export of ER (With NER)								
1	400 kV	BINAGURI-BONGAIGAON	2	72	243	0.1	2.8	-2.7
2	400 kV	ALIPURDUAR-BONGAIGAON	2	150	531	0.0	5.3	-5.3
3	220 kV	ALIPURDUAR-SALAKATI	2	18	101	0.0	1.3	-1.3
ER-NER						0.1	9.4	-9.3
Import/Export of NER (With NR)								
1	HVDC	BISWANATH CHARIALI-AGRA	2	0	503	0.0	12.4	-12.4
NER-NR						0.0	12.4	-12.4
Import/Export of WR (With NR)								
1	HVDC	CHAMPA-KURUKSHETRA	2	0	5029	0.0	92.2	-92.2
2	HVDC	VINDHYACHAL B/B	-	0	309	0.0	7.0	-7.0
3	HVDC	MUNDRA-MOHINDERGARH	2	0	2357	0.0	49.0	-49.0
4	765 kV	GWALIOR-AGRA	2	0	2986	0.0	43.6	-43.6
5	765 kV	GWALIOR-PHAGI	2	552	1581	1.2	19.4	-18.1
6	765 kV	JABALPUR-ORAI	2	0	1489	0.0	45.8	-45.8
7	765 kV	GWALIOR-ORAI	1	606	0	10.8	0.0	10.8
8	765 kV	SATNA-ORAI	1	0	1099	0.0	20.4	-20.4
9	765 kV	BANASKANTHA-CHITORGARH	2	704	2076	1.6	17.5	-15.9
10	765 kV	VINDHYACHAL-VARANASI	2	0	3939	0.0	70.4	-70.4
11	400 kV	ZERDA-KANKROLI	1	195	364	1.1	2.7	-1.6
12	400 kV	ZERDA -BHINMAL	1	135	251	0.8	1.9	-1.1
13	400 kV	VINDHYACHAL -RIHAND	1	977	0	21.9	0.0	21.9
14	400 kV	RAPP-SHUJALPUR	2	239	703	0.9	6.2	-5.3
15	400 kV	NEEMUCH-Chittorgarh	2	287	656	1.1	5.8	-4.6
16	220 kV	BHANPURA-RANPUR	1	0	110	0.0	2.1	-2.1
17	220 kV	BHANPURA-MORAK	1	0	30	0.0	2.1	-2.1
18	220 kV	MEHGAON-AURAIYA	1	73	24	0.6	0.0	0.6
19	220 kV	MALANPUR-AURAIYA	1	43	43	0.2	0.1	0.0
20	132 kV	GWALIOR-SAWAI MADHOPUR	1	0	0	0.0	0.0	0.0
21	132 kV	RAJGHAT-LALITPUR	2	0	0	0.0	0.0	0.0
WR-NR						40.2	385.9	-345.7
Import/Export of WR (With SR)								
1	HVDC	BHADRAWATI B/B	-	998	0	23.9	0.0	23.9
2	HVDC	RAIGARH-PUGALUR	2	479	0	11.6	0.0	11.6
3	765 kV	SOLAPUR-RAICHUR	2	1710	880	22.0	0.8	21.2
4	765 kV	WARDHA-NIZAMABAD	2	185	2600	0.1	21.1	-21.0
5	765 kV	WARORA-WARANGAL(NEW)	2	215	2373	0.3	18.2	-17.9
6	400 kV	KOLHAPUR-KUDGI	2	1888	0	36.2	0.0	36.2
7	220 kV	KOLHAPUR-CHIKODI	2	0	0	0.0	0.0	0.0
8	220 kV	PONDA-AMBEWADI	1	0	0	0.0	0.0	0.0
9	220 kV	XELDEM-AMBEWADI	1	1	111	1.5	0.0	1.5
WR-SR						95.4	40.1	55.3

INTERNATIONAL EXCHANGES							Import(+ve)/Export(-ve)
State	Region	Line Name	Max (MW)	Min (MW)	Avg (MW)	Energy Exchange (MU)	
BHUTAN	ER	400kV MANGDECHHU-ALIPURDUAR 1,2&3 i.e. ALIPURDUAR RECEIPT (from MANGDECHU HEP 4*180MW)	737	691	712	17.09	
	ER	400kV TALA-BINAGURI 1,2,4 (& 400kV MALBASE - BINAGURI) i.e. BINAGURI RECEIPT (from TALA HEP 6*170MW)	935	839	849	20.39	
	ER	220kV CHUKHA-BIRPARA 1&2 (& 220kV MALBASE - BIRPARA) i.e. BIRPARA RECEIPT (from CHUKHA HEP 4*84MW)	148	83	96	2.30	
	NER	132kV GELEPHU-SALAKATI	37	27	33	0.79	
	NER	132kV MOTANGA-RANGIA	69	38	58	1.38	
NEPAL	NR	NEPAL IMPORT (FROM UP)	0	0	0	0.00	
	NR	132kV MAHENDRANAGAR-TANAKPUR(NHPC)	70	0	56	1.34	
	ER	NEPAL IMPORT (FROM BIHAR)	0	0	0	0.00	
	ER	400kV DHALKEBAR-MUZAFFARPUR 1&2	723	580	612	14.69	
BANGLADESH	ER	BHERAMARA B/B HVDC (B'DESH)	-926	-924	-925	-22.20	
	ER (Isolated from Indian Grid)	400kV GODDA_TPS-RAHANPUR (B'DESH) D/C	-1288	-802	-927	-22.25	
	NER	132kV COMILLA-SURAJMANI NAGAR 1&2	-84	0	-71	-1.70	

Export From India (in MU)

Country	GNA (ISGS/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Nepal	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Bangladesh	22.16	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.16
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Export	22.16	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.16

Import by India(in MU)

Country	GNA (ISGA/PPA)	T-GNA							TOTAL	
		BILATERAL TOTAL	COLLECTIVE							
			IDAM			RTM				
			IEX	PXIL	HPX	IEX	PXIL	HPX		
Bhutan	38.76	0.00	1.12	0.00	0.00	0.00	0.00	0.00	39.88	
Nepal	2.63	0.00	13.32	0.00	0.00	0.00	0.16	0.00	0.00	16.11
Bangladesh	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Import	41.39	0.00	14.44	0.00	0.00	0.00	0.16	0.00	0.00	55.99

Net from India(in MU)

-ve : Export / +ve : Import

Country	GNA (ISGS/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	38.76	0.00	1.12	0.00	0.00	0.00	0.00	0.00	39.88
Nepal	2.63	0.00	13.32	0.00	0.00	0.16	0.00	0.00	16.11
Bangladesh	-22.16	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-22.16
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Net	19.23	0.00	14.44	0.00	0.00	0.16	0.00	0.00	33.83